

ARTICLES

# Beech Leaf Disease

This article discusses the cause of beech leaf disease (BLD), the lifecycle of this nematode (*Litylenchus crenatae* ssp. *mccannii*), the symptoms of BLD, and treatments.

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Updated: March 6, 2024



## Introduction

Beech Leaf Disease (BLD) is a substantial threat to American beech (*Fagus grandifolia*) and all other species of beech (*Fagus* spp.) across the eastern United States. American beech plays an important role in forest ecosystems; as a late successional species it adds structural diversity to a variety of forest types and its nut production makes it an excellent species for wildlife.

Beech leaf disease was first identified in Lake County, Ohio, in 2012, but has since been found across the eastern United States. BLD is easily to recognizes by its distinctive dark green interveinal banding pattern on symptomatic leaves. It is caused by *Litylenchus crenatae* ssp. *mccannii* (LCM), a nematode that parasites tree leaves. This damage is seen in most species of beech. It can be more difficult to identify early in beech with dark leaves like "copper" beech (*Fagus sylvatica* 'Purpurea').



Figure 1. A beech midstory in Pennsylvania that is being attacked by *Litylenchus crenatae* ssp. *mccannii*. Photo credit: Calvin Norman

## Symptoms

The dark bands in leaves become evident in early spring as leaves emerge fully symptomatic (Figure 1). Under high levels of infection, leaf symptoms become more pronounced; leaves become leathery and crinkled. These bands are due to cellular damage (Figure 2) caused by nematodes in the leaf buds (Figure 3). In forests heavily impacted by BLD, it may look like there is a drought that is impacting only beech.



>Figure 2. A cellular view of a bud damaged by nematodes. The healthy bud cells are those that are smaller (a) and flat next to the very large and disordered cells (b) that have been impacted by nematodes. Photo credit: Mihail Kantor

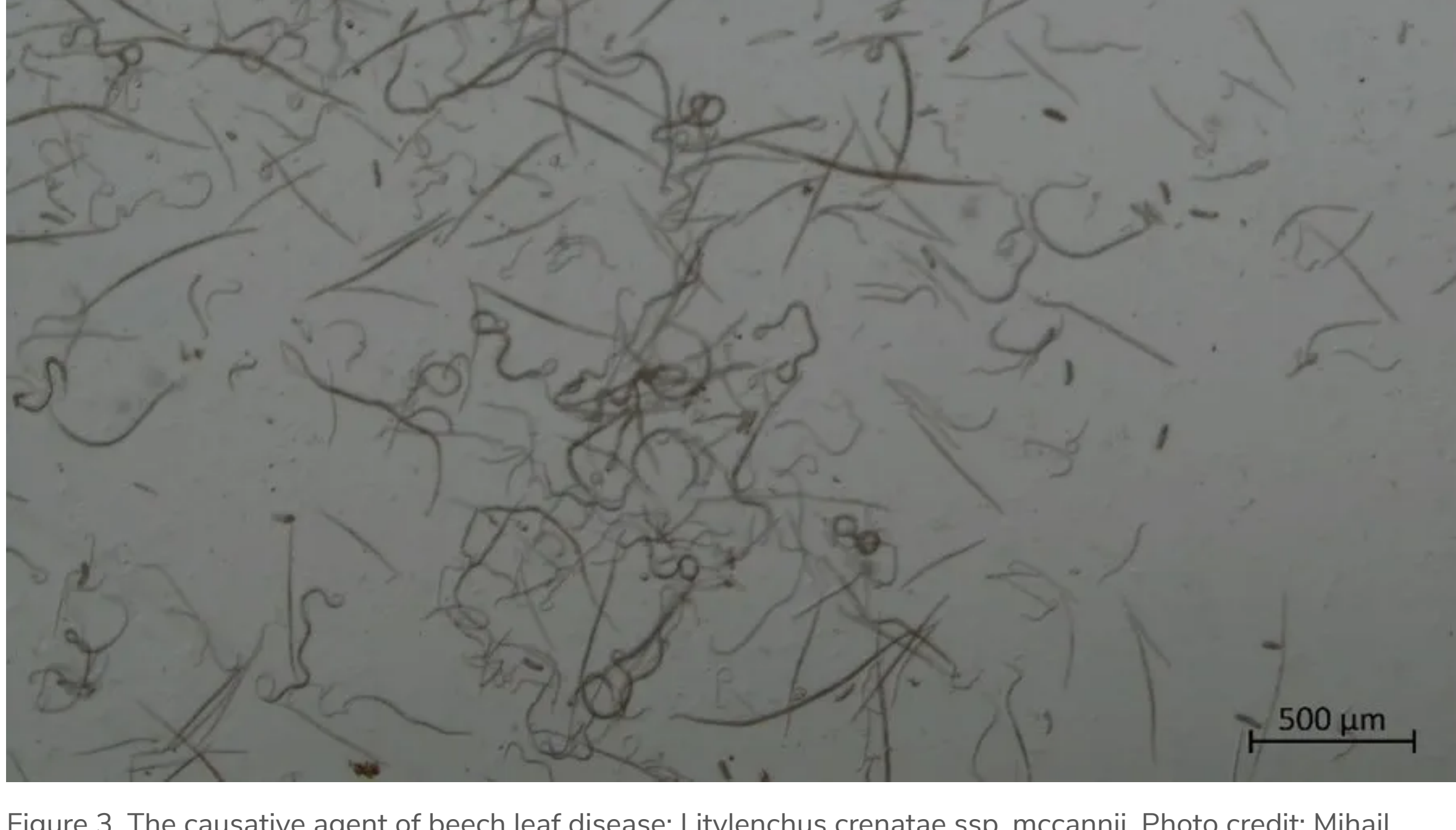


Figure 3. The causative agent of beech leaf disease: *Litylenchus crenatae* ssp. *mccannii*. Photo credit: Mihail Kantor

## Disease Progression

In spring, from May to June, there are few nematodes in diseased leaves. The number of nematodes increases over the summer (July–August) and reaches a peak in late summer through early fall (September–November). As the numbers build up within the leaves, nematodes will migrate from the infected leaves into the newly forming buds. Once nematodes reach the new bud, they will feed on the bud tissues, which causes cell enlargement of the bud scale cells and negatively affects the overall architecture of the developing young leaves by increasing their number of cell layers (Figure 4). This damages the next generation of leaves and repeated damage will kill infected beech. Currently, the method of transmission between infected trees is unknown.

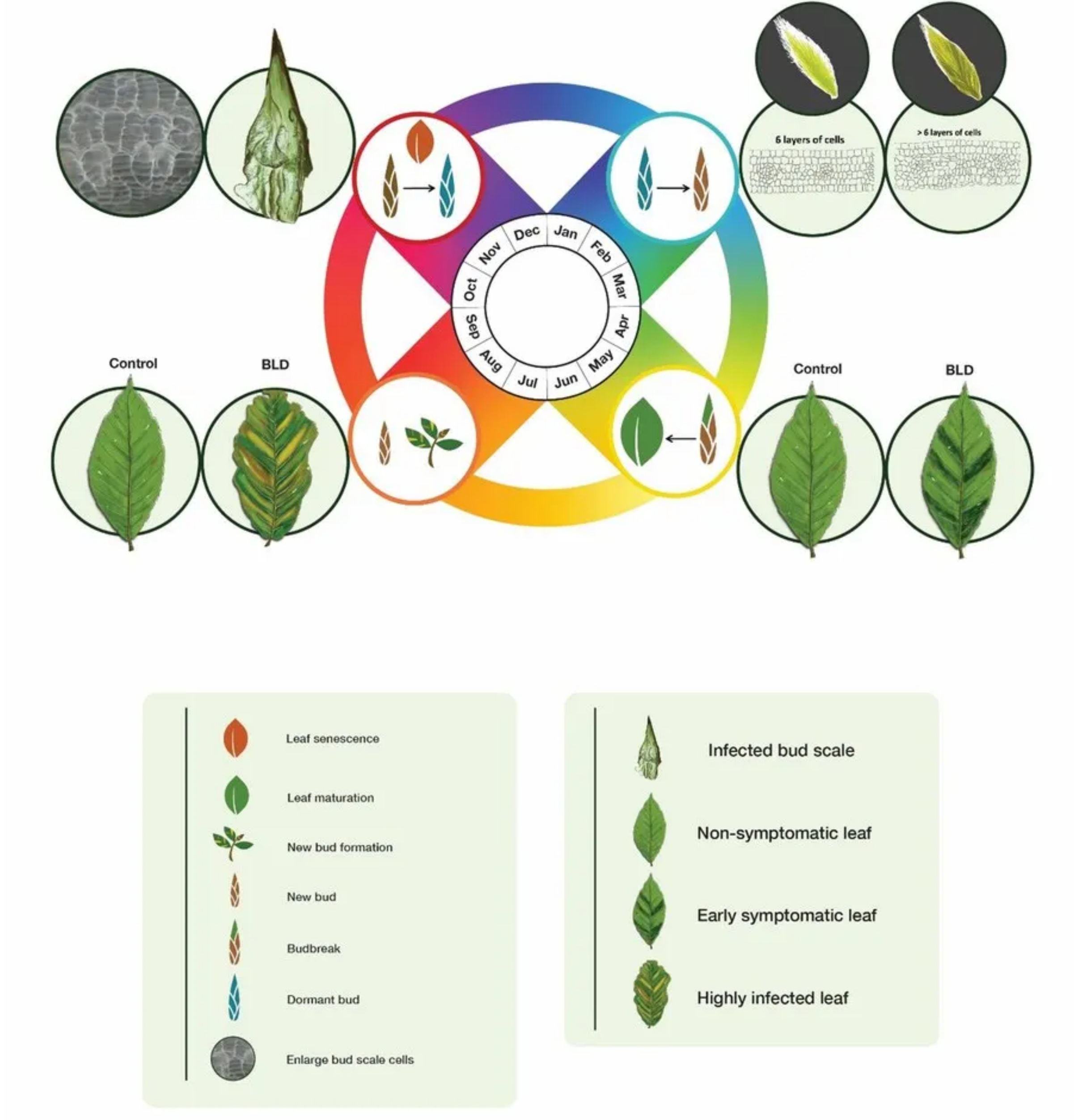


Figure 4. The progression of beech leaf disease. The graphic illustration was conceptualized by Mihail Kantor and Paulo Viera. It was created by Michael Tribone and Parisa Farjamfar.

## Treatment

### Yard/Street Trees

BLD can be treated in yard trees by injections of thiabendazole (like Arbotect 20-S). According to the label, inject 2-8 fl oz of thiabendazole for every 5 inches of trunk diameter and dilute with 16-32 parts water for one part chemical. For large trees (stems with a DBH of 8 inches or more), stems may be injected with up to 12 fl oz thiabendazole per 5 inches of trunk diameter. Always read and understand the chemical label prior to applying. Other products that show promising results against BLD are potassium phosphite, potassium polyphosphate, and other potassium fertilizers. The recommended application rate is 2 fl. oz. of phosphite products mixed with 14 oz. of water for every inch of diameter and applied at the base of the infected tree. Trees that have been damaged for multiple years may not be saved as the damage is too much for the trees to overcome. Long term research is still underway.

### Management in Forests

Currently there is no feasible large-scale treatment for BLD in forests. Individual trees can be treated using the treatments for individual trees, but this can be expensive.

## Conclusion

Beech leaf disease is a disease that impacts American beech and several other species of beech. It is caused by a nematode, *Litylenchus crenatae* ssp. *mccannii*. It damages leaf buds which leads to leaf loss, canopy thinning, and aborted buds. Beech leaf disease has killed a large number of beech in the eastern United States and is increasing in both the severity and spread. In yard trees, there are several treatment methods, however, it should be noted that these treatments are expensive and research is still underway. In a forest setting, this treatment is expensive and typically not feasible. With beech leaf disease and beech bark disease on the landscape, American beech is imperiled.

### Authors

**Calvin Norman**  
Assistant Teaching Professor  
of Forestry

#### Expertise

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**Mihail Kantor**  
Assistant Research Professor

Penn State University

[mpk6148@psu.edu](mailto:mpk6148@psu.edu)



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